

Linux module: 2

Assignment

Que: View running process with ps.

Ans:

1) Basic ps command: To view the processes running in the current terminal session:

Command: ps

2) List all processes: To list all running processes for all users on the system:

Command: ps -e

3) Show processes with detailed information: To display a more detailed list of all processes, including PID, memory usage, CPU time, etc.

Command: ps aux

4) Show processes with specific user: To show processes for a specific user:

Command: ps -u username

5) Show hierarchical process tree: To display processes in a tree format, showing parent-child relationship:

Command: ps axjf

Que: Terminate process with kill.

Ans:

- 1) Find the process ID (PID): First, you need to identify the processes you want to terminate. Use `ps`, `top`, or `grep` to find the PID.
- 2) Send a termination signal: The default signal is `SIGTERM` (signal number 15), which gracefully asks the process to terminate.
- 3) Kill multiple processes: You can kill multiple processes at once by passing multiple PIDs to the `kill` command.
- 4) Kill a process by name: if you want to terminate a process by its name, you can combine `pgrep` with `kill`.

Que: Use top to monitor system resources and processes.

Ans:

- 1) Start top: To start monitoring system resources, simply run the `top` command. Once `top` is running, it will display the system's CPU usage, memory usage, process list and other information.
- 2) Understanding the top output:
 - Summary area: This provides high-level information about the system.
 - Uptime: How long the system has been running.
 - Load average: The system load over the last 1, 5 and 15 minutes.

- CPU usage: The percentage of CPU time used by user processes, system processes and idle time.
- Memory usage: The total physical memory and how much is used, free, or buffered.

3) Basic command inside top:

- Quit top: press q to exit top.
- Change the update interval: You can change the refresh rate by pressing d, then entering the number of seconds between updates.
- Search for a specific process: press / and type the process name or part of it, then press enter. Top will highlight the matching processes.
- Kill a process: press k, then type the PID of the process you want to terminate. It will prompt you to confirm the signal to send.

4) Advanced options for top:

- Display all processes: To show processes from all users, use: top -u <username>
- Show system resources summary only: To get a summary of system resources without the full process list, use the -d option to specify a delay and -b for batch mode.
- Interactive search and help: Press h to get a list of available commands while top is running.

Que: Configure one of your lab computers to boot to the CLI using system and reboot to confirm that you were successfully.

Ans: done in lab.